CE1O1 – CONTINUUM LIMITS FROM SECOND-MOMENT BOUNDS

Abstract. A uniform second-moment estimate for finite linear combinations of lattice fields gives subsequential continuum laws after cell rescaling. This working outline isolates the general theorem, its measurable realization and compactness arguments, and its application to the already constructed infinite-volume QED3 laws. Each box links to a full statement and a source-based proof budget.

Contents

1. Introduction and target statement 1
Proof overview 3
2. E. Realizing and estimating the step field 4
3. K. Second moments imply compactness 5
4. G. The supplied infinite-lattice QED3 estimate 6
5. Q. A subsequential continuum law for QED3 7

1. Introduction and target statement

The volume limit is taken first at each fixed spacing. CEM1, Theorem B and Sections 3–4, now prove the general existence argument. Historical IV1M1 locators refer to the preserved real-curvature revision in sources/manifest.json; current IVM1 treats compact links. LPM1 supplies the current real-curvature laws and Gaussian bound. The criterion applies to finitely many real components in any finite dimension.

T. Continuum limits of lattice fields

Interface. Overview Target.

Theorem A. Fix integers $d, r \geq 1$, offsets $t_1, \dots, t_r \in [0,1]^d$, and a set $I \subset (0,1]$ with 0 in its closure. For each $a \in I$, let ν_a be a Borel probability law of real arrays $u = (u_\alpha(x))_{x \in \mathbb{Z}^d, 1 \leq \alpha \leq r}$, with the coordinate product topology. Suppose one constant $C < \infty$ satisfies

$$\mathbb{E}_{\nu_a} \left| \sum_{\alpha,x} b_\alpha(x) u_\alpha(x) \right|^2 \leq C \sum_{\alpha,x} |b_\alpha(x)|^2 \tag{1}$$

for every $a \in I$ and every finitely supported real array b. Define the physical step field by

$$(F_a)_\alpha(y) = a^{-d/2} u_\alpha(x), \qquad y \in Q_{a,\alpha,x} := a(x + t_\alpha + [-1/2, 1/2)^d). \tag{2}$$

Then F_a is almost surely a real tempered distribution.

For the normalized scalar Hermite functions h_k on $\mathbb{R}^d$, put $\lambda_k = 2|k|_1 + d + 1$ and

$$H_{-t} = \left\{ F \in \mathcal{S}'(\mathbb{R}^d; \mathbb{R}^r) : \|F\|_{-t}^2 := \sum_{\alpha=1}^{r} \sum_{k \in \mathbb{N}_0^d} \lambda_k^{-2t} |F(h_k e_\alpha)|^2 < \infty \right\}.$$

Here e_α is the standard basis of $\mathbb{R}^r$ and $|k|_1 = k_1 + \cdots + k_d$. For every fixed $t > d/2$, the fields F_a have Borel laws on the separable Hilbert space H_{-t}, and these laws are uniformly tight in its

norm topology. Consequently, every sequence $a_j \in I$ with $a_j \to 0$ has a subsequence a_{j_ℓ} and a Borel probability law μ on H_{-t} such that

$$\mathcal{L}(F_{a_{j_\ell}}) \implies \mu.$$

In particular, the limit is a law of tempered distributions; pairings with every finite family of Schwartz tests converge jointly in distribution along this subsequence. ■□

The estimate concerns all finite linear combinations of lattice coordinates. A bound on each coordinate separately would not supply the uniform estimate for smeared fields used here. No compatibility between the laws at different spacings is assumed. The conclusion is subsequential existence; it does not assert uniqueness or that the limit is nonzero.

Proof to write. First apply E to construct the step distribution and prove $\mathbb{E}|F_a(f)|^2 \le C\|f\|_2^2$ for every real Schwartz test. Then apply K to this family of distributions. Choose any s with $d/2 < s < t$ in the compactness argument. The normalization $a^{-d/2}$ is chosen so that cell averaging preserves the quadratic bound.

Relation to the existing outline. The compactness and recovery module B3 in the preserved whole-proof outline QE1O3 includes finite tori, fixed-spacing recovery and convergence of moments. The present theorem extracts the existence argument after the infinite-volume law has been obtained. The general criterion needs only (1); QED3 supplies it with $C = 1$ through G. The application is Q. The current proof is CEM1, Theorem B and Sections 3–4. In the historical source locators below, IV1M1 denotes the earlier real-curvature revision preserved in this outline's `sources/manifest.json`; current IVM1 treats compact links.

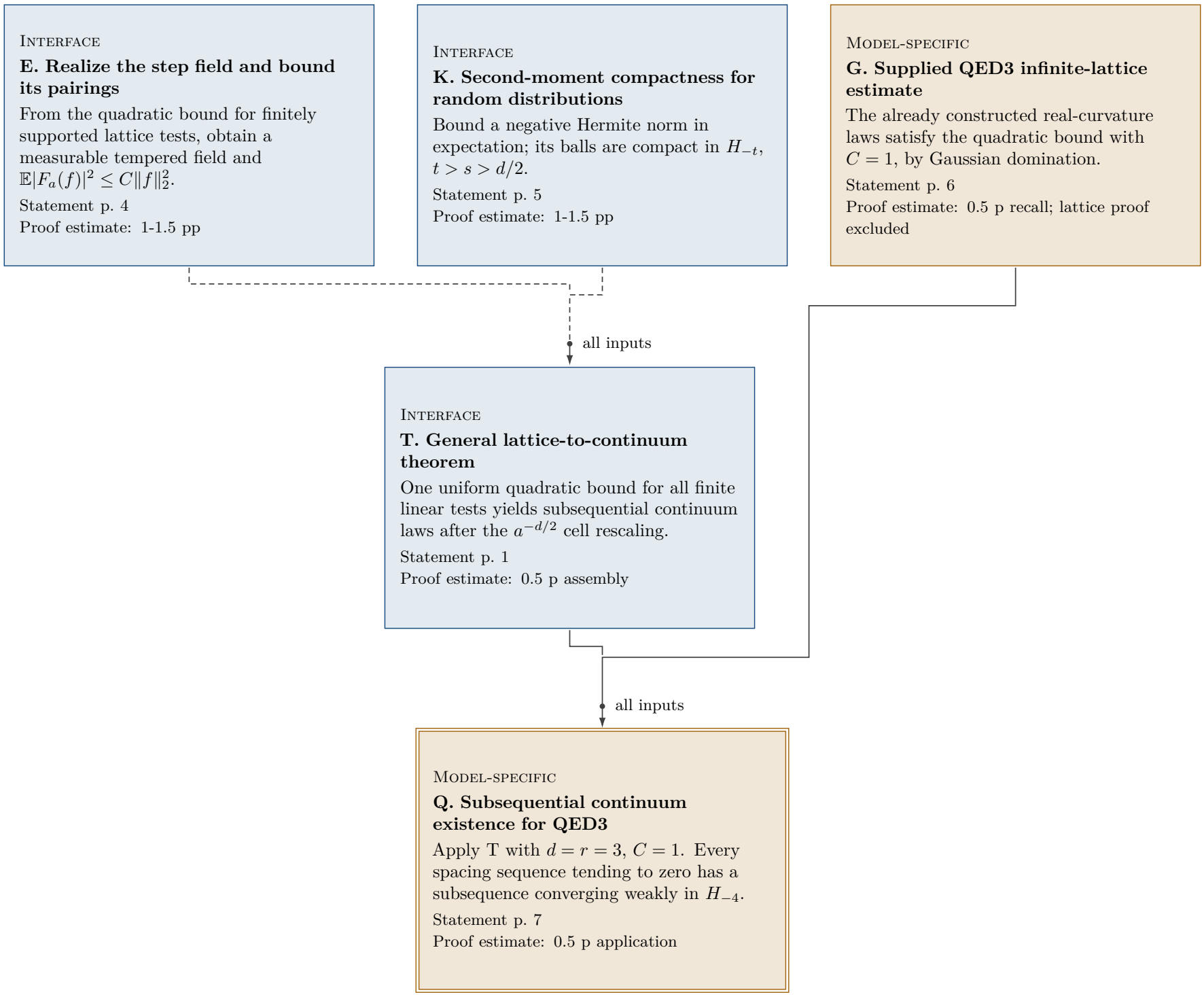
Source. QE1M1, §5.4 and §6.2, pp. 52 and 54; TeX 4394–4422 and 4551–4582. The infinite-lattice realization and unfolded test estimate are written in the preserved IV1M1 revision, `real-fields.tex`, lines 121–171, and `continuum.tex`, lines 17–72. Sources are preserved with this outline; their manifest records the exact revisions.

Proof estimate: $\frac{1}{2}$ page for the assembly, in addition to E and K. The source already combines these steps in the first half of QE1M1 p. 54.

CE1O1 – Continuum limits from second-moment bounds

Definitions/setup | Model-specific | Interface (general)

CEO1 is the focused compactness outline for CEM1 Theorem B. T is the general theorem; Q is its QED3 application, using the real-curvature input from LPM1.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

Standing input for E and T: genuine infinite-lattice probability laws with a common second-moment bound for every finite real linear test. All full hypotheses are on the linked pages. Proof budgets exclude upstream lattice results; the extracted write-up is about 4-6 pages.

2. E. Realizing and estimating the step field

Interface. Overview Target.

Statement. Fix $d, r \geq 1$, $0 < a \leq 1$, and $t_\alpha \in [0,1]^d$. Let ν be a Borel law of real arrays on $\mathbb{Z}^d \times \{1, \dots, r\}$ satisfying

$$\mathbb{E}_\nu \left| \sum_{\alpha,x} b_\alpha(x) u_\alpha(x) \right|^2 \leq C \|b\|_{\ell^2}^2 \quad \text{for every finitely supported real } b.$$

Put $Q_{a,\alpha,x} = a(x + t_\alpha + [-1/2, 1/2)^d)$ and let $(F_a)_\alpha = a^{-d/2} u_\alpha(x)$ on that cube. This defines a tempered distribution on a measurable set of full ν measure. Its pairing with each real $f \in \mathcal{S}(\mathbb{R}^d; \mathbb{R}^r)$ is the almost surely absolutely convergent sum

$$F_a(f) = \sum_{\alpha,x} b_{a,\alpha}(x) u_\alpha(x), \qquad b_{a,\alpha}(x) = a^{-d/2} \int_{Q_{a,\alpha,x}} f_\alpha(y)\, dy.$$

The pairing is measurable, its finite truncations converge in $L^2(\nu)$, and

$$\mathbb{E}_\nu |F_a(f)|^2 \leq C \|b_a\|_{\ell^2}^2 \leq C \|f\|_{L^2}^2. \tag{3}$$

Both constants are independent of a, the offsets, and the law.

What requires proof. An L^2 limit of the pairing for each individual test does not by itself identify one samplewise tempered distribution. The weighted lattice estimate supplies that realization. Choose $p > d/2$ and define $\|u\|_{\ell^2_{-p}}^2 = \sum_{\alpha,x} (1 + |x|^2)^{-p} |u_\alpha(x)|^2$. The coordinate instances of the hypothesis give

$$\mathbb{E}_\nu \|u\|_{\ell^2_{-p}}^2 \leq Cr \sum_{x \in \mathbb{Z}^d} (1 + |x|^2)^{-p} < \infty.$$

Thus this norm is finite on one measurable set of full measure. For every array in that set, weighted Cauchy–Schwarz bounds the absolute pairing sum by

$$\sum_{\alpha,x} |b_{a,\alpha}(x) u_\alpha(x)| \leq \|u\|_{\ell^2_{-p}} \left(\sum_{\alpha,x} (1 + |x|^2)^p |b_{a,\alpha}(x)|^2 \right)^{1/2}. \tag{4}$$

For fixed a, the second factor is bounded by a Schwartz seminorm of f. This proves continuity in the test variable and hence temperedness. The constant in this realization estimate may depend on a; uniformity is required in (3) instead. Finite coordinate sums are measurable, so their pointwise limits are measurable. Define the field to be zero on the exceptional set.

For each component, its cubes partition $\mathbb{R}^d$. Cauchy–Schwarz on a cube, followed by summation, gives

$$\sum_{\alpha,x} \left| a^{-d/2} \int_{Q_{a,\alpha,x}} f_\alpha \right|^2 \leq \sum_{\alpha,x} \int_{Q_{a,\alpha,x}} |f_\alpha|^2 = \|f\|_2^2.$$

Apply the assumed quadratic bound to differences of finite truncations of b_a. They form an $L^2(\nu)$ Cauchy sequence; its limit agrees with the samplewise sum in (4). This passes the quadratic inequality to $F_a(f)$ and proves (3).

Source. QE1M1, §6.1, pp. 52–53, especially (6.1), (6.3) and (6.4); §6.4, p. 56, for measurable step evaluation. The second-moment realization used here is the weighted argument in the preserved IV1M1 revision, `real-fields.tex`, lines 121–133 and 157–171; the passage to infinite coefficient sums is in `continuum.tex`, lines 17–45. These arguments apply componentwise; the cell volume is a^d.

Proof estimate: 1–1$\frac{1}{2}$ pages. The relevant QE1M1 material occupies parts of pp. 52–53 and 56; the weighted realization is supplied by that earlier revision and is now proved in CEM1, Proposition 3.1. Do not count the finite-volume inverse map as part of this proof.

3. K. Second moments imply compactness

Interface. Overview Target.

Statement. Let $d, r \geq 1$, and let $(X_i)_{i\in J}$ be any family of random real tempered distributions on $\mathbb{R}^d$ with r components. Assume that $X_i(f)$ is measurable for each real Schwartz test and that one constant $C < \infty$ satisfies

$$\mathbb{E}|X_i(f)|^2 \leq C\|f\|^2_{L^2(\mathbb{R}^d;\mathbb{R}^r)} \qquad (i \in J,\ f \text{ real Schwartz}).$$

Use the normalized Hermite basis $h_k e_\alpha$ of $L^2(\mathbb{R}^d;\mathbb{R}^r)$, with $\lambda_k = 2|k|_1 + d + 1$, and define H_{-s} by the norm

$$\|X\|^2_{-s} = \sum_{\alpha=1}^{r} \sum_{k\in\mathbb{N}_0^d} \lambda_k^{-2s} |X(h_k e_\alpha)|^2.$$

For $s > d/2$ the fields admit Borel realizations in H_{-s} and

$$\sup_i \mathbb{E}\|X_i\|^2_{-s} \leq B_s := Cr \sum_{k\in\mathbb{N}_0^d} \lambda_k^{-2s} < \infty. \tag{5}$$

For every $t > s$ and $R > 0$, the set $K_R = \{X \in H_{-s} : \|X\|_{-s} \leq R\}$ is compact in H_{-t} and

$$\inf_i \mathbb{P}(X_i \in K_R) \geq 1 - B_s/R^2.$$

The laws are therefore uniformly tight on H_{-t}, and every sequence has a weakly convergent subsequence there. The inclusion $H_{-t} \hookrightarrow \mathcal{S}'(\mathbb{R}^d;\mathbb{R}^r)$ is continuous for the strong distribution topology.

The compactness argument. The Hermite coefficients convert the test estimate into a bound on a norm of the whole field. Tonelli gives (5). Its sum converges because the number of indices with $|k|_1 = n$ is $\binom{n+d-1}{d-1}$, of order $(1+n)^{d-1}$. The coefficients are measurable; their finite Hermite sums converge in H_{-s} almost surely. This realizes the field as a Borel random variable in a separable Hilbert space. The limit agrees with the given tempered distribution by its Hermite expansion on Schwartz tests.

The loss from s to t produces the compact set. On K_R,

$$\sum_{\alpha,k:\,\lambda_k > M} \lambda_k^{-2t} |X(h_k e_\alpha)|^2 \leq M^{-2(t-s)} R^2. \tag{6}$$

The remaining coefficients range over a bounded subset of a finite-dimensional space. Select a convergent subsequence on these finite sets and use (6) to control the tail uniformly. Fatou's lemma keeps the limit in K_R. Markov's inequality now gives the displayed probability bound; Prokhorov's theorem gives weak subsequential limits on H_{-t}.

Pairing with a Schwartz test is continuous by weighted Cauchy–Schwarz in Hermite coordinates. The corresponding positive Hermite norm of a test is bounded by a Schwartz seminorm, uniformly on bounded sets of tests. This also verifies the asserted continuous inclusion in the strong distribution topology.

The conclusion is convergence in law; the hypotheses alone do not give convergence of second or higher moments.

Source. QE1M1, §5.4, p. 52, and the Hermite–Tonelli estimate (6.11), p. 54; the preserved IV1M1 revision, `continuum.tex`, lines 47–72. The current compactness proof is CEM1, Proposition 4.1. Hermite background: R. Radha and S. Thangavelu, *Holomorphic Sobolev spaces, Hermite and special Hermite semigroups*, §§2,4,5. Probability compactness: E. Sorensen, *TA Session on tightness of probability measures*, Theorem 1.1 and Example 2.2.

Proof estimate: 1–1$\frac{1}{2}$ pages, including measurability and dimension-dependent summability; based on the compact-ball and expected-norm arguments on QE1M1 pp. 52,54.

4. G. The supplied infinite-lattice QED3 estimate

Model-specific. Overview Target.

Input from the lattice construction. At each spacing a, take one of the already constructed infinite-volume laws ν_a of the real chamber-QED3 curvature array $u_{ij}(x)$, $x \in \mathbb{Z}^3$, $1 \le i < j \le 3$. The precise output needed from that construction is

$$\mathbb{E}_{\nu_a} \exp\left(\sum_{x,i<j} b_{ij}(x)u_{ij}(x)\right) \le \exp\left(\tfrac{1}{2}\sum_{x,i<j} |b_{ij}(x)|^2\right) \tag{7}$$

for every finitely supported real b. The constant is independent of spacing and of the selected volume limit. In particular,

$$\mathbb{E}_{\nu_a}\left|\sum_{x,i<j} b_{ij}(x)u_{ij}(x)\right|^2 \le \sum_{x,i<j} |b_{ij}(x)|^2. \tag{8}$$

This is the sole model-specific estimate used in the continuum existence argument.

What is already available. LPM1, Theorem A(i), constructs the current real-curvature volume-limit laws; Proposition 5.1 gives finite-volume Gaussian domination. Its parameters are fixed during each volume limit. At a fixed spacing and for a fixed finite support, sufficiently large tori contain that support without overlap. Weak convergence of the lattice laws passes the bounded truncations of the exponential; monotone convergence then gives (7). CEM1, Section 7, records this passage using LPM1. The historical sources are QE1M1, (6.16), and the infinite-volume theorem of the preserved IV1M1 revision. The quadratic consequence (8) follows from the supplied random-linear-form argument in QE1M1, Proposition 5.1: apply the exponential estimate to tb and compare the first two derivatives at $t = 0$ for both signs of t.

The infinite-volume law is a law of real curvature. A law of compact link variables alone is not the input in this statement. The finite-volume construction and its Gaussian estimate are upstream results; this outline does not repeat their proofs.

Source. QE1M1, Theorem 3.5 (`thm:slab-gaussian`); Proposition 5.1 (`prop:subgaussian`), pp. 49–50; and (6.16), p. 56. The preserved IV1M1 revision, `real-fields.tex`, `thm:qed-main`, and `continuum.tex`, lines 17–45. The master allows a real bare-mass assignment depending on the spacing and independent of outer volume: QE1M1, §6.2, p. 54.

Proof estimate: $\frac{1}{2}$ page to recall the input and deduce its quadratic consequence. The finite-volume Gaussian proof and infinite-volume construction are already separate results and are excluded from this budget.

5. Q. A subsequential continuum law for QED3

Model-specific. Overview Target.

Statement. Let $a_n \to 0$ be a sequence of positive spacings. For each n, choose an infinite-volume real-curvature law ν_{a_n} on arrays $u_{ij}(x)$, $x \in \mathbb{Z}^3$, $1 \le i < j \le 3$, satisfying

$$\mathbb{E}_{\nu_{a_n}} e^{\sum_{x,i<j} b_{ij}(x)u_{ij}(x)} \le e^{\frac{1}{2}\sum_{x,i<j}|b_{ij}(x)|^2} \quad (b \text{ finitely supported and real}).$$

On the cube $a_n(x + (e_i + e_j)/2 + [-1/2, 1/2)^3)$ put $(F_{a_n})_{ij} = a_n^{-3/2} u_{ij}(x)$. There is a subsequence whose laws converge weakly in H_{-4} to a Borel probability law of real tempered two-forms on $\mathbb{R}^3$. Here

$$\|F\|_{-4}^2 = \sum_{i<j} \sum_{k \in \mathbb{N}_0^3} (2|k|_1 + 4)^{-8} |F(h_k\, dy_i \wedge dy_j)|^2,$$

where h_k are the normalized scalar Hermite functions. The already constructed QED3 laws satisfy the required input by G.

Proof to write. Use G to verify the hypothesis of theorem A with $d = r = 3$ and $C = 1$. The component offsets are $(e_i + e_j)/2$. Apply T with $t = 4$, using $s = 3$ for the compact balls. Each spacing may use any selected infinite-volume law; the bound is common to all choices.

What this completes. This supplies a continuum probability law from the real lattice laws. Reflection positivity, covariance, field equations, clustering and interaction are outside this compactness statement. CEM1, Corollary 7.1, proves the stated existence conclusion and additional Gaussian and correlation conclusions. Its Theorem C and Corollary 7.2 address closedness and the supplied symmetries and reflection forms under their additional hypotheses. Nine-family reflection positivity is not an input to this compactness step.

Size of the extracted proof. The core write-up, including the general statement, E, K and this application, should occupy about 4–6 ordinary manuscript pages. This is an editorial estimate from the cited portions of the master, allowing room for the measurable realization. The pages in this linked outline contain separate statement cards and are not a proposed manuscript page count. The budgets exclude the existing lattice construction, Gaussian-domination proof and proofs of the standard compactness background.

Source. QE1M1, §5.4 and §6.2, pp. 52 and 54, together with the infinite-lattice identification in §6.4, pp. 55–56. The direct route from an already infinite-volume real law is the preserved IV1M1 revision, `continuum.tex`, lines 17–78.

Proof estimate: $\frac{1}{2}$ page for this application, excluding T, E, K and the recalled input G.